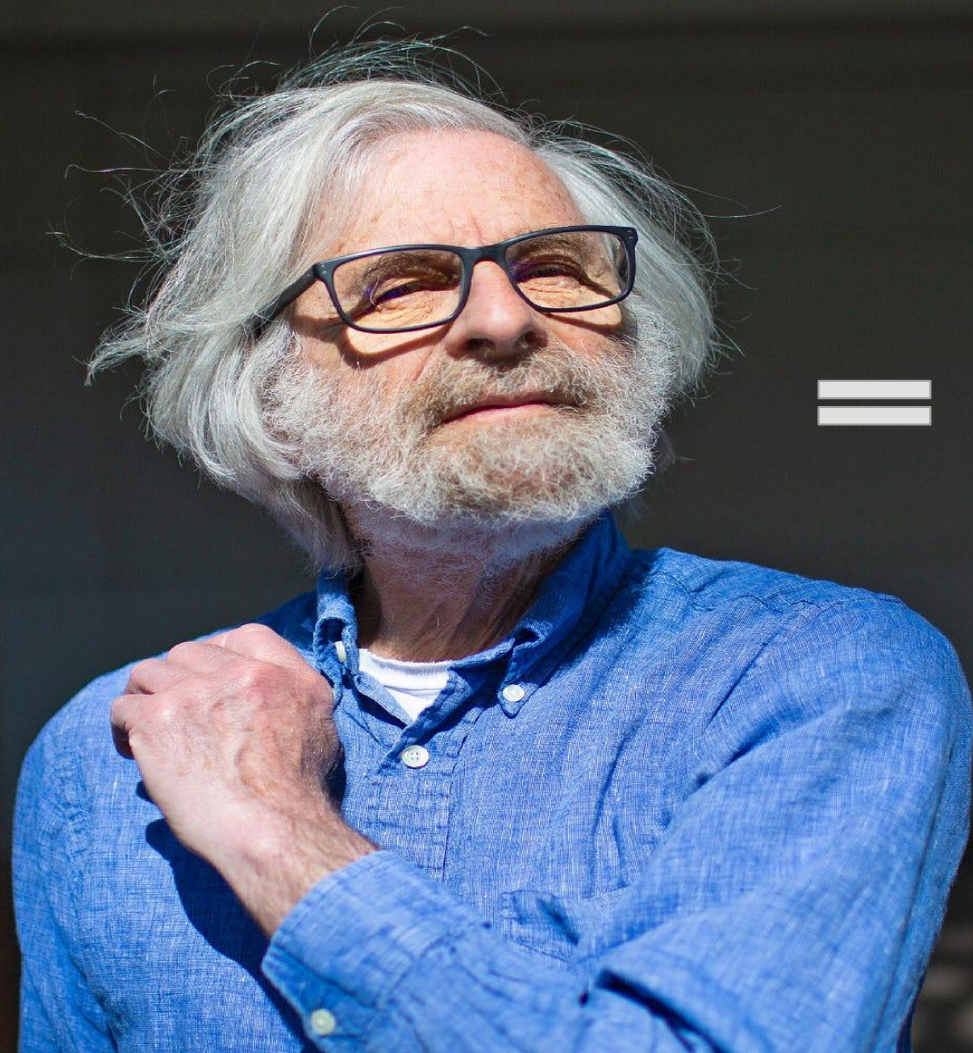


Introduction to LaTeX

Pronounced as LAY-tek



Lecture 1



=



The Man Who Revolutionized Computer Science With Math



Formatting is no
substitute for writing.

Leslie Lamport

quote fancy

Courtesy of quotefancy

Please take **five** minutes to reflect on your thoughts. ✍️



L^AT_EX

What is LaTeX?

- A tool used for creating professional-looking documents
- Especially useful for academic and technical writing

Compared to Word Processors

- More control over formatting
- Less visual editing, more coding
- Better output for complex content (math, references, layouts)



Why Use LaTeX?

- Ideal for project reports, STEM assignments, and research papers
- Handles:
 - **Math equations** with precision
 - **Citations** using BibTeX or BibLaTeX
 - **Tables and figures** with consistent formatting
 - **Large documents** with structured sections

Introduction to Overleaf

- Overleaf is an **online** LaTeX editor.
- No installation needed — works directly in your **browser** with real-time preview.
- Ideal for students working on project reports, research papers, lab reports, theses, and group projects.
- Offers templates, version control, and **GitHub** integration.
- Supports collaboration, allowing multiple users to edit the same document simultaneously.

LaTeX and Overleaf are not interchangeable!!



[Log in to Overleaf - Overleaf, Online LaTeX Editor](#)

A meme featuring the character Elmo from Sesame Street. He is shown from the chest up, with his arms raised in a 'V' shape. He has a surprised or excited expression with wide eyes and an open mouth. He is surrounded by intense, bright orange and yellow flames that fill the background. The overall tone is humorous and urgent.

GAMETIME LETS LOG IN!!!

Writing your first piece of LaTeX

```
\documentclass{article}
```

report, book, CV...

```
\begin{document}
```

First document. This is a simple example, with no extra parameters or packages included.

```
\end{document}
```

First document. This is a simple example, with no extra parameters or packages included.

What are Preambles in LaTeX?

- Appears before **`\begin{document}`**
- Used to:
 - Set up document formatting
 - Create the title page
 - Load packages
- Controls how your document looks and behaves
- You can customize:
 - Font size and style
 - Line spacing
 - Page margins

```
\documentclass[12pt,a4paper]{report}
\usepackage[utf8]{inputenc}
\usepackage{geometry}
\usepackage{graphicx}
\usepackage{longtable}
\usepackage{array}
\usepackage{enumitem}
\usepackage{hyperref}
\usepackage{fancyhdr}
\usepackage{titlesec}
\usepackage{setspace}
\usepackage{listings}
\geometry{margin=1in}
```

Adding Comments

```
\documentclass[12pt, letterpaper]{article}
\title{My first LaTeX document}
\author{Danish Puri\thanks{Funded by the Liberty team.}}
\date{June 2025}
\begin{document}
\maketitle
We have now added a title, author and date to our first \LaTeX{} document!

% This line here is a comment. It will not be typeset in the document.
\end{document}
```



Bold, *italics* and underlining

- **Bold**: bold text in LaTeX is typeset using the `\textbf{...}` command.
- *Italics*: italicised text is produced using the `\textit{...}` command.
- Underline: to underline text use the `\underline{...}` command.

```
Some of the \textbf{greatest}  
discoveries in \underline{science}  
were made by \textbf{\textit{accident}}.
```

Some of the **greatest** discoveries
in science were made by *accident*.

List : itemized

```
\documentclass{article}
\begin{document}
\begin{itemize}
  \item The individual entries are indicated with a black dot, a so-called bullet.
  \item The text in the entries may be of any length.
\end{itemize}
\end{document}
```

- The individual entries are indicated with a black dot, a so-called bullet.
- The text in the entries may be of any length

List : numbered

```
\documentclass{article}
\begin{document}
\begin{enumerate}
  \item This is the first entry in our list.
  \item The list numbers increase with each entry we add.
\end{enumerate}
\end{document}
```

1. This is the first entry in our list
2. The list number increase with each entry we add

Tables

Table `\ref{table:data}` shows how to add a a table caption and reference a table.

```
\begin{table}[h!]
```

```
\centering
```

```
\caption{Table to test captions and labels.}
```

```
\label{table:data}
```

```
\begin{tabular}{||c c c c||}
```

```
\hline
```

```
Col1 & Col2 & Col2 & Col3 \\\ [0.5ex]
```

```
\hline\hline
```

```
1 & 6 & 87837 & 787 \\\
```

```
2 & 7 & 78 & 5415 \\\
```

```
3 & 545 & 778 & 7507 \\\
```

```
4 & 545 & 18744 & 7560 \\\
```

```
5 & 88 & 788 & 6344 \\\ [1ex]
```

```
\hline
```

```
\end{tabular}
```

```
\end{table}
```

`\\` denotes ends of a row.

`&` denotes end of a column

`\hline` inserts horizontal line between two rows

Table 1 shows how to add a table caption and reference a table.

Table 1: Table to test captions and labels.

Col1	Col2	Col2	Col3
1	6	87837	787
2	7	78	5415
3	545	778	7507
4	545	18744	7560
5	88	788	6344

Images

```
\begin{figure}[h]
  \centering
  \includegraphics[width=0.75\textwidth]{mesh}
  \caption{A nice plot.}
  \label{fig:mesh1}
\end{figure}
```

As you can see in figure `\ref{fig:mesh1}`, the function grows near the origin. This example is on page `\pageref{fig:mesh1}`.

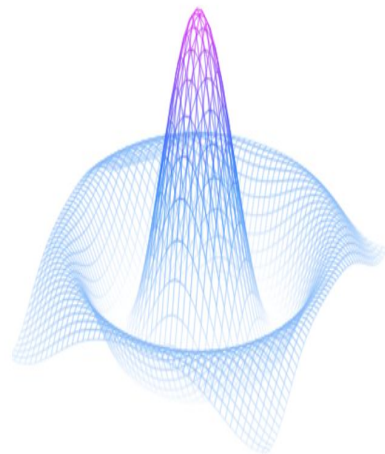


Figure 1: A nice plot.

As you can see in figure 1, the function grows near the origin. This example is on page 1.

Assignment

- Make **title page** of your project report in LaTeX and send the overleaf link to the department email.
- Due: **Tuesday, 24 June at 11:59 p.m.**